

# Vibrational Wiring Deduction Transformer (VDT):

A Spectral-PPCA Generative Framework for  
Graph-Structured Latent Spaces and Spectral Associative Memory

Lorenzo Moriondo

Independent Researcher — [tuned.org.uk](https://tuned.org.uk)

ORCID: 0000-0002-8804-2963

[github repo](#)

June 2026

## Abstract

We present the *Vibrational Wiring Deduction Transformer* (VDT), a fully Bayesian generative model that unifies vibrational graph dynamics, probabilistic PCA in a Laplacian eigenbasis (Spectral-PPCA), and spectral associative memory into a single trainable architecture. The predecessor Wiring Autoencoder (WAE) used a hard high-frequency energy penalty ( $J_{\text{freq}}$ ) and an isotropic Gaussian latent prior. VDT replaces both with principled variational KL terms: (i) a spectral-basis KL over the loading matrix  $\mathbf{S}$  weighted by Laplacian eigenvalues, implementing a *spectral Occam's razor*; (ii) a closed-form Gamma/Exponential KL over per-mode weights  $\omega$ , providing soft  $\tau$ -mode band-limiting; and (iii) the standard isotropic Gaussian latent KL retained for stability (the Laplacian-precision latent KL is preserved as a design option and ablation, but is not included in the default three-term ELBO following architectural review in PR #35). After training, a *spectral artefact*  $\mathcal{A}(\mathcal{I})$  is extracted and used to initialise a downstream transformer with pre-built spectral associative memory. The two-phase (offline training, online inference) architecture separates spectral learning from deployment, with frozen eigenpairs eliminating all runtime  $O(N^3)$  eigendecompositions during training. We detail the complete module design, three-term ELBO, stability analysis, benchmark metrics, and associative memory construction.

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# 1 Introduction

Graph-structured latent variable models occupy an increasingly important niche between classical variational autoencoders (VAEs) and geometric deep learning: they can encode both the *content* of nodes and the *relational structure* that organises them. The Wiring Autoencoder [?] recognised this, using a differentiable graph Laplacian  $L(z)$  as an intermediate latent object whose spectral properties are regularised by a hard frequency penalty  $J_{\text{freq}} = \sum_{j>k} \lambda_j(L(z))$ . While effective in practice, this penalty is a heuristic rather than a proper probabilistic prior, which makes it difficult to:

1. compare competing ArrowSpace indices  $\mathcal{I}_1, \mathcal{I}_2$  via principled model selection;
2. interpret the learned spectral structure as a posterior over loading matrices  $\mathbf{W}$ ;
3. extract a reusable *spectral artefact* whose keys are approximately orthonormal and whose values are grounded in the data manifold.

The VDT addresses all three limitations by embedding the Wiring Autoencoder within the Spectral-PPCA framework [Moriondo, 2026a], a Bayesian generative model in which the loading matrix  $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$  lives explicitly in the Laplacian eigenbasis and is endowed with eigenvalue-weighted shrinkage priors. The VDT encoder backbone implements a discrete damped wave equation in the graph spectral domain, connecting the probabilistic latent space to the physical intuition of vibrational modes in string and spring systems.

## 1.1 Contributions

- A three-term ELBO replacing  $J_{\text{freq}}$  and preserving the isotropic latent prior (Section 4).
- A **SpectralLoadingDecoder** that maps latent codes to eigenbasis-factorised loading matrices (Section 5).
- A **VibrationalEncoder** (VDT recurrence) with **ModeWeightHead** producing variational Gamma parameters (Section 5).
- A **SpectralAssociativeMemory** module that pre-builds a Hopfield memory from the spectral artefact (Section 6).
- A two-phase offline/online architecture eliminating runtime eigensolves during training (Section 7).
- An eight-metric evaluation suite including ELBO Bayes factors for ArrowSpace index selection (Section 9).

# 2 Background

## 2.1 Probabilistic PCA

Standard PPCA [Tipping and Bishop, 1999] defines a linear-Gaussian generative model:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_q), \quad \mathbf{x} \mid \mathbf{z} \sim \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2 \mathbf{I}_d), \quad (1)$$

producing marginal  $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{W}\mathbf{W}^\top + \sigma^2 \mathbf{I}_d)$ . Maximum-likelihood estimation recovers the PCA loading matrix in the column space of the  $q$  largest singular vectors of the data matrix. The fully Bayesian extension places priors over  $\mathbf{W}$ ,  $\mathbf{z}$ , and  $\sigma^2$ , enabling model selection over the latent dimensionality  $q$  via the marginal likelihood [Minka, 2001].

## 2.2 Graph Laplacian Spectral Theory

Let  $G = (V, E, w)$  be a weighted graph with node set  $|V| = N$ . The normalised graph Laplacian is  $L = D^{-1/2}(D - W)D^{-1/2}$  where  $D_{ii} = \sum_j w_{ij}$ . Eigendecomposition  $L = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$  yields eigenvectors  $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N) \in \mathbb{R}^{N \times N}$  and non-negative eigenvalues  $0 \leq \lambda_1 \leq \dots \leq \lambda_N$ . Low-frequency eigenvectors (small  $\lambda_k$ ) are smooth over  $G$ ; high-frequency ones (large  $\lambda_k$ ) oscillate rapidly across edges [Chung, 1997].

Given ArrowSpace index  $\mathcal{I}$ , we denote the frozen  $q$ -truncated eigendecomposition

$$L(\mathcal{I}) = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top, \quad \mathbf{U}_q = \mathbf{U}_{:,1:q} \in \mathbb{R}^{N \times q}, \quad \mathbf{\Lambda}_q = \text{diag}(\lambda_1, \dots, \lambda_q). \quad (2)$$

These eigenpairs are computed once offline and used as frozen constants throughout training.

## 2.3 Vibrational Dynamics and Spectral Graph Waves

The VDT encoder draws its physical motivation from Rayleigh’s Theory of Sound [?]: the eigenmodes of a vibrating string or spring system are exactly the eigenvectors of the system’s stiffness matrix, which in the discrete graph setting is the normalised Laplacian  $L$ . A positive amplitude on mode  $k$  corresponds to a string displacement in the direction  $\mathbf{u}_k$ ; a negative amplitude corresponds to the opposing displacement. This is the physical counterpart to the signed (positive/negative) latent amplitudes that the VDT encoder learns.

The signed density matrix  $\varrho_t = \varrho_t^+ - \varrho_t^-$  maintained by the VDT recurrence (Section 5.2) is precisely a decomposition of the current state into constructive and destructive vibrational contributions. The low-entropy state of this system corresponds to coherent oscillation (the string vibrating in a definite mode); the high-entropy state corresponds to rest (the wave function collapsed, all modes equally weighted). This mirrors Brownian motion, where the low-entropy state is the non-heated stationary system and the high-entropy state is maximal thermal agitation.

## 2.4 Wiring Autoencoder (WAE) — Predecessor

WAE [?] trains a differentiable Laplacian  $L(\mathbf{z})$  whose topology is fixed (kNN + RBF graph) but whose edge weights are learned by a mixture-of-experts decoder  $W_{\text{dec}} : \mathbf{z} \mapsto \delta_e$ . The training objective is

$$\mathcal{L}_{\text{WAE}} = \mathbb{E}_q[\log p(\mathbf{x} \mid \mathbf{z})] - \beta \cdot \text{KL}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) - \alpha \cdot J_{\text{freq}}(L(\mathbf{z})), \quad (3)$$

where  $J_{\text{freq}} = \sum_{j>k} \lambda_j(L(\mathbf{z}))$  penalises high-frequency wiring. The reconstruction is provided by **TauModeDiffusion**: the heat kernel  $K_\tau = \mathbf{U}_k \exp(-t\mathbf{\Lambda}_k)\mathbf{U}_k^\top$  applied to an embedding table  $\mathbf{E}$ , with learnable diffusion time  $t$ .

Three limitations of WAE motivate VDT: (a)  $J_{\text{freq}}$  is a deterministic penalty with no prior interpretation; (b) the latent prior is isotropic and ignores graph geometry; (c) there is no principled post-training artefact for downstream use.

## 3 Spectral-PPCA Generative Framework

Following Moriondo [2026a], we define the Spectral-PPCA prior by replacing the standard PPCA loading matrix with one that lives in the Laplacian eigenbasis.

### 3.1 Three Structural Priors

**Prior 1: Spectral-basis loading.** Parametrise  $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$  where  $\mathbf{U}_{1:q} \in \mathbb{R}^{d \times q}$  are the  $q$  lowest-frequency eigenvectors of  $L(\mathcal{I})$  (frozen) and  $\mathbf{S} \in \mathbb{R}^{q \times q}$  lives in the spectral basis. Place an eigenvalue-weighted Gaussian shrinkage prior on  $\mathbf{S}$ :

$$p(\mathbf{S} \mid \mathcal{I}) \propto \exp\left(-\frac{\lambda_s}{2} \text{tr}(\mathbf{S}^\top \mathbf{\Lambda}_q \mathbf{S})\right). \quad (4)$$

This implements a *spectral Occam’s razor*: high-frequency components (large  $\lambda_k$ ) are penalised exponentially more strongly than low-frequency ones, pulling the loading matrix toward smooth directions of the index graph.

**Prior 2:  $\tau$ -mode frequency prior.** Let  $\omega_k > 0$  be a weight on the  $k$ -th mode. The  $\tau$ -mode prior is a product of exponentials with rates that scale with Laplacian eigenvalues:

$$p(\omega_k \mid \tau, \lambda_k) = \text{Exp}(\tau \lambda_k), \quad k = 1, \dots, q. \quad (5)$$

Small  $\lambda_k$  (low-frequency / low-entropy vibrational modes) receive weak penalisation; large  $\lambda_k$  receive strong penalisation, naturally selecting the effective spectral bandwidth. The loading matrix is  $\mathbf{W} = \mathbf{U}_{1:q} \text{diag}(\boldsymbol{\omega}) \mathbf{S}$ .

**Prior 3 (design option): Laplacian-precision latent prior.** For a richer latent geometry one can replace the isotropic prior with a Laplacian-precision prior:

$$p(\mathbf{z} \mid \mathcal{I}) = \mathcal{N}(\mathbf{0}, (\mathbf{I} + \beta L_s)^{-1}), \quad (6)$$

where  $L_s$  is a sample-graph Laplacian built from the current batch of latent means (stop-gradient). The resulting KL expands to

$$\text{KL}(q(\mathbf{z}) \parallel p_{\text{Lap}}(\mathbf{z})) = \frac{1}{2} \left[ \text{tr}(\Sigma_z (\mathbf{I} + \beta L_s)) + \boldsymbol{\mu}_z^\top (\mathbf{I} + \beta L_s) \boldsymbol{\mu}_z - d - \log \det \Sigma_z + \log \det (\mathbf{I} + \beta L_s)^{-1} \right], \quad (7)$$

where the cross-term  $\boldsymbol{\mu}_z^\top L_s \boldsymbol{\mu}_z$  is the graph Dirichlet energy of the latent means. *This prior is preserved in the VDT codebase as an ablation option (`use_isotropic_kl=False` in `VibrationalEncoder`) but is **not** included in the default three-term ELBO following architectural review in PR #35.*

## 4 VDT: Three-Term ELBO

### 4.1 Variational Family

We adopt a mean-field amortised variational family:

$$q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_z, \text{diag}(\exp(\mathbf{s}_z))), \quad (8)$$

$$q_\phi(\mathbf{S} \mid \mathbf{x}) = \mathcal{N}(M_S, \text{diag}(\exp(\mathbf{v}_S))), \quad (9)$$

$$q_\phi(\omega_k \mid \mathbf{x}) = \text{Gamma}(a_k, b_k), \quad (10)$$

where all parameters are outputs of the `VibrationalEncoder` network. The Gamma distribution is reparametrisable via the implicit reparametrisation trick [Figurnov et al., 2018]; in practice we parametrise through  $(\log a_k, \log b_k)$  from a `ModeWeightHead` linear layer.

### 4.2 ELBO Decomposition

The default VDT training objective is the following three-term ELBO (minimisation form, as implemented in `vdt/model.py`):

$$\mathcal{L}_{\text{VDT}} = \underbrace{\mathbb{E}_q[\|\mathbf{x} - \hat{\mathbf{x}}\|^2 / 2\sigma^2]}_{\text{reconstruction}} + \underbrace{\text{KL}(q(\mathbf{z}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))}_{\text{isotropic latent KL (kl\_z)}} + \underbrace{\text{KL}(q(\mathbf{S}) \parallel p(\mathbf{S} \mid \mathcal{I}))}_{\text{spectral basis KL (kl\_S)}} + \underbrace{\text{KL}(q(\boldsymbol{\omega}) \parallel p(\boldsymbol{\omega} \mid \tau, \boldsymbol{\Lambda}))}_{\tau\text{-mode KL (kl\_tau)}} \quad (11)$$

All four terms are non-negative; the total loss is minimised (ELBO is maximised by negating) during gradient descent.

### 4.3 Closed-Form KL Terms

**Isotropic latent KL.**

$$\text{KL}(q(\mathbf{z})\|\mathcal{N}(\mathbf{0}, \mathbf{I})) = -\frac{1}{2} \sum_{i=1}^{d_z} (1 + s_{z,i} - \mu_{z,i}^2 - e^{s_{z,i}}). \quad (12)$$

**Spectral-basis KL.** For the entry  $(k, j)$  of  $\mathbf{S}$ , with precision  $\lambda_s \lambda_k$  (eq. (4)):

$$\text{KL}_{kj} = \frac{1}{2} [\lambda_s \lambda_k (v_{S,kj} + S_{kj}^2) - v_{S,kj} - 1 - \log(\lambda_s \lambda_k)], \quad (13)$$

where  $v_{S,kj} = \exp(\log \text{var}_{S,kj})$ . In the current implementation the log-variance is approximated as  $\log \text{var}_{S,kj} = \log(S_{kj}^2 + \epsilon)$ , deriving uncertainty from the magnitude of the mean — a simplification that avoids adding a separate posterior variance head to `SpectralLoadingDecoder`. Future work should add a dedicated variance head for a proper amortised posterior.

**$\tau$ -mode KL.** For mode  $k$  with  $q(\omega_k) = \text{Gamma}(a_k, b_k)$  and  $p(\omega_k) = \text{Gamma}(1, \tau \lambda_k)$ , the closed-form KL is:

$$\text{KL}(\text{Gamma}(a, b)\|\text{Exp}(r)) = \log b - \log r + \ln \Gamma(a) + (1 - a)\psi(a) + \frac{ab}{r}, \quad r = \tau \lambda_k, \quad (14)$$

where  $\psi$  is the digamma function. This is summed over all  $q$  modes and averaged over the batch.

### 4.4 Comparison with WAE Predecessor

Table 1: ELBO comparison: WAE (predecessor) vs VDT.

| Term                     | WAE                                                             | VDT                                                                            | Note                         |
|--------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|
| Reconstruction           | $-\ \mathbf{x} - \hat{\mathbf{x}}\ ^2 / 2\sigma^2$              | Same                                                                           | $\sigma$ learnable           |
| Latent KL                | $\text{KL}(q(\mathbf{z})\ \mathcal{N}(\mathbf{0}, \mathbf{I}))$ | Same (isotropic)                                                               | PR #35 decision              |
| Spectral basis KL        | —                                                               | $\text{KL}(q(\mathbf{S})\ p(\mathbf{S}   \mathcal{I}))$                        | eigenvalue-weighted Gaussian |
| $\tau$ -mode term        | $\alpha J_{\text{freq}}$ (hard)                                 | $\text{KL}(q(\boldsymbol{\omega})\ p(\boldsymbol{\omega}   \tau, \mathbf{A}))$ | variational Gamma prior      |
| Lap. precision latent KL | —                                                               | <i>design option</i>                                                           | ablation only                |

## 5 Model Architecture

### 5.1 Data Flow

Figure 1 illustrates the end-to-end data flow.

### 5.2 VibrationalEncoder

`VibrationalEncoder` (`vdt/encoder.py`) replaces the WAE MLP encoder with a VDT (Vibrational Discrete Transformer) recurrence stack:

1. **Input projection:** linear layer mapping  $(B, D) \rightarrow (B, N \cdot d)$ , reshaped to  $(B, N, d)$ .
2. **VibrationalStateBlocks:**  $K$  recurrence layers operating on  $(B, N, d)$  state matrices using the feature-space Laplacian  $L_f$ . The recurrence implements a discrete damped wave equation in the graph spectral domain: the state evolves as  $\mathbf{h}_{t+1} = (2\mathbf{I} - \alpha L_f)\mathbf{h}_t - \mathbf{h}_{t-1}$ , where  $\alpha$  is a learnable damping coefficient. This is a finite-difference discretisation of the wave equation  $\ddot{\mathbf{u}} + \alpha L_f \mathbf{u} = 0$ .

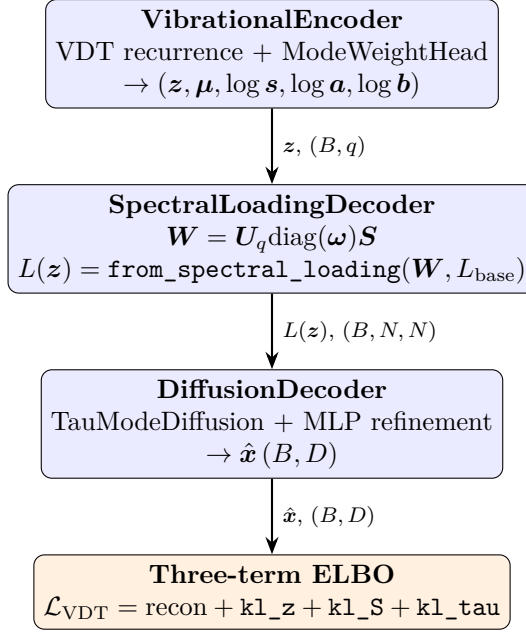


Figure 1: VDT data flow. The eigenpairs  $(\mathbf{U}_q, \mathbf{\Lambda}_q)$  of  $L(\mathcal{I})$  are frozen constants passed to both encoder and decoder; no Laplacian is computed at runtime.

3. **Modal projection:** mean-pool output over the node dimension after projecting through eigenvectors  $\mathbf{v} \in \mathbb{R}^d$ .
4. **Readout heads:** separate linear layers producing  $\boldsymbol{\mu}, \log \mathbf{s}$  (for  $q(\mathbf{z})$ ) and **ModeWeightHead** producing  $(\log \mathbf{a}, \log \mathbf{b})$  for  $q(\boldsymbol{\omega})$ .

The optional  $\lambda$ -fingerprint (ArrowSpace spectral histogram of  $L(\mathcal{I})$ ) can be concatenated to  $\mathbf{x}$  before the input projection (`use_lambda_features=True`).

**ModeWeightHead.** A single linear layer `Linear(feats_dim, 2q)` splits into  $(\log \mathbf{a}, \log \mathbf{b}) \in \mathbb{R}^q \times \mathbb{R}^q$ . The Gamma parameters are obtained as  $a_k = e^{\log a_k} > 0$ ,  $b_k = e^{\log b_k} > 0$ . The mode-weight KL (14) is computed from these in `vdtspectral.py:tau_mode_kl`.

**Signed density matrix.** During the VDT recurrence each block maintains a signed density matrix  $\varrho_t = \varrho_t^+ - \varrho_t^-$ , decomposing the state into constructive and destructive vibrational contributions (positive and negative amplitudes). This object is the discrete analogue of the quantum density matrix when amplitudes are real-valued rather than complex. It is used as a stability diagnostic and, optionally, as the seed for the spectral artefact in reasoning-focused deployments (Option 6, Section 10).

### 5.3 SpectralLoadingDecoder

`SpectralLoadingDecoder` (`vdts/wiring_decoder.py`) is the default VDT wiring decoder. The forward pass computes:

$$\mathbf{S} = \mathbf{S\_net}(\mathbf{z}).\text{view}(B, q, q), \quad (15)$$

$$\boldsymbol{\omega} = \exp(\boldsymbol{\omega\_net}(\mathbf{z})), \quad (16)$$

$$\mathbf{W} = \mathbf{U}_q.\text{unsqueeze}(0) @ (\boldsymbol{\omega}.\text{unsqueeze}(-1) * \mathbf{S}), \quad (17)$$

$$L(\mathbf{z}) = \text{DifferentiableLaplacian.from\_spectral\_loading}(\mathbf{W}, L_{\text{base}}). \quad (18)$$

Edge weights in (18) are synthesised as  $w_{ij} = w_{ij}^{\text{base}} \cdot \sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2)$  where  $\mathbf{W}_i$  is the  $i$ -th row of  $\mathbf{W}$  and  $\sigma$  is the logistic sigmoid. This path is fully differentiable from  $L(\mathbf{z})$  back to  $\mathbf{z}$  through  $\mathbf{W}$ .

**Initialisation.** `S_net.weight` is initialised so the initial  $\mathbf{S}_b \approx \mathbf{I}_q$  (near-identity); `omega_net` is initialised near zero so initial  $\omega_k \approx 1$  for all modes. This gives a warm-start close to the eigenbasis reparametrisation  $\mathbf{W} \approx \mathbf{U}_q$ .

**Config dispatch.** The legacy WAE `WiringDecoder` (mixture-of-experts) is retained as a fallback: `decoder_type: spectral` selects `SpectralLoadingDecoder`; `decoder_type: mixture_of_experts` selects `WiringDecoder`.

## 5.4 DiffusionDecoder

`DiffusionDecoder` is retained from vdt. It wraps `TauModeDiffusion`: the truncated heat kernel

$$K_\tau = \mathbf{U}_k \text{diag}(e^{-t\Lambda_k}) \mathbf{U}_k^\top, \quad (19)$$

with learnable log-diffusion-time  $\log t$ . A small MLP refinement residually corrects the output. Reconstruction loss is Gaussian NLL:  $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 / (2\sigma^2)$ .

The heat kernel (19) has a direct vibrational interpretation:  $e^{-t\lambda_k}$  is the amplitude decay factor for mode  $k$  after time  $t$ , exactly as in the damped wave equation. Modes with large  $\lambda_k$  (high-frequency, high-entropy) decay faster; low-frequency modes persist.

## 5.5 VibrationalDeductionTransformer (Top-level Module)

`VibrationalDeductionTransformer` (vdt/model.py) assembles all three modules. Its `forward()` method returns a dict with exactly nine keys: `{loss, recon, kl_z, kl_S, kl_tau, x_hat, z, mu, log_var}`.

**Model factory.** `from_config(cfg, E)` dispatches on `cfg['model']['version']`: version 1 builds the legacy `WiringAutoencoder`; version 2 builds `VibrationalDeductionTransformer`.

# 6 Spectral Artefact and Associative Memory

## 6.1 Spectral Artefact

After training to convergence, the spectral artefact  $\mathcal{A}(\mathcal{I})$  is extracted by evaluating the model at the prior mean  $\mathbf{z} = \mathbf{0}$ :

$$\mathcal{A}(\mathcal{I}) = \left\{ \hat{\mathbf{W}} = \mathbf{U}_q \text{diag}(\hat{\boldsymbol{\omega}}) \hat{\mathbf{S}}, \quad \hat{\boldsymbol{\omega}} = (\hat{\omega}_1, \dots, \hat{\omega}_q), \quad S_{\mathcal{I}} = \sum_{k=1}^q \hat{\omega}_k \hat{\mathbf{w}}_k \hat{\mathbf{w}}_k^\top \right\}, \quad (20)$$

where  $\hat{\mathbf{w}}_k$  is the  $k$ -th column of  $\hat{\mathbf{W}}$ . Equation (20) uses  $d_\theta(\hat{\mathbf{w}}_k) \approx \hat{\mathbf{w}}_k$  (linear decoder approximation for  $S_{\mathcal{I}}$ ); a full nonlinear value function  $d_\theta(\hat{\mathbf{w}}_k)$  is left as future work (Section 12).

## 6.2 SpectralAssociativeMemory

`SpectralAssociativeMemory` (vdt/spectral\_memory.py) wraps  $S_{\mathcal{I}}$  as a pre-built Hopfield / linear associative memory. For query  $\mathbf{q} \in \mathbb{R}^{d_k}$ , retrieval proceeds via:

$$\mathbf{o} = \text{softmax}\left(\beta_H S_{\mathcal{I}}^\top \mathbf{q}\right) S_{\mathcal{I}}, \quad (21)$$

where  $\beta_H$  is the inverse temperature controlling retrieval sharpness. The approximately-orthonormal keys  $\hat{\mathbf{w}}_k$  (columns of  $\hat{\mathbf{W}}$  are linear combinations of eigenvectors, orthonormal when  $\mathbf{S} \approx \mathbf{I}$ ) maximise retrieval SNR [Ramsauer et al., 2021]:  $\text{SNR}^{-1} \approx N_{\text{stored}}/d_k$ .

Table 2: Components of the spectral artefact  $\mathcal{A}(\mathcal{I})$ .

| Component                 | Symbol                                         | Meaning                                     |
|---------------------------|------------------------------------------------|---------------------------------------------|
| Posterior loading basis   | $\hat{\mathbf{W}} \in \mathbb{R}^{d \times q}$ | Mean loading matrix in Laplacian eigenbasis |
| Posterior mode weights    | $\hat{\boldsymbol{\omega}} \in \mathbb{R}^q$   | Per-mode activity under $\tau$ -mode prior  |
| Associative memory matrix | $S_{\mathcal{I}} \in \mathbb{R}^{d \times d}$  | Outer-product Hopfield key matrix           |

**Vibrational interpretation.** The columns of  $\hat{\mathbf{W}}$  are superpositions of Laplacian eigenmodes: low-frequency modes (small  $\lambda_k$ , high  $\hat{\omega}_k$ ) correspond to the fundamental vibrational tones of the graph, while high-frequency modes correspond to harmonics that are suppressed by the  $\tau$ -mode prior. The memory matrix  $S_{\mathcal{I}}$  thus stores information in the basis of the system’s natural resonances — an associative memory grounded in physical intuition.

**Online delta-rule updates.** New associations can be written without overwriting spectral structure:

$$S_{\mathcal{I}} \leftarrow S_{\mathcal{I}} + \frac{1}{B}(\mathbf{v} - S_{\mathcal{I}}\mathbf{k})\mathbf{k}^{\top}, \quad (22)$$

where  $(\mathbf{k}, \mathbf{v})$  are new key-value pairs.

## 7 Two-Phase Architecture

VDT cleanly separates spectral learning from online inference.

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### Algorithm 1 VDT Two-Phase Procedure

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**Require:** ArrowSpace index  $\mathcal{I}$ , dataset  $D$ , hyperparameters

```

1: // Phase 1 — Offline spectral learning
2: Compute eigendecomposition:  $(\mathbf{U}_q, \boldsymbol{\Lambda}_q) \leftarrow \text{eigh}(L(\mathcal{I}))$  ▷ once;  $O(N^3)$ 
3: Freeze  $(\mathbf{U}_q, \boldsymbol{\Lambda}_q)$ 
4: for each training step do
5:    $\mathbf{z}, \boldsymbol{\mu}, \log \mathbf{s}, \log \mathbf{a}, \log \mathbf{b} \leftarrow \text{VibrationalEncoder}(\mathbf{x})$ 
6:    $\mathbf{W}, \boldsymbol{\omega}, \mathbf{S}, L(\mathbf{z}) \leftarrow \text{SpectralLoadingDecoder}(\mathbf{z}, \mathbf{U}_q, L_{\text{base}})$ 
7:    $\hat{\mathbf{x}} \leftarrow \text{DiffusionDecoder}(L(\mathbf{z}), \mathbf{E})$ 
8:   Compute  $\mathcal{L}_{\text{VDT}}$  (eq. (11)) and update  $\phi, \theta$ 
9: end for
10:  $\mathcal{A}(\mathcal{I}) \leftarrow \text{extract\_spectral\_artefact}(\mathbf{U}_q, \boldsymbol{\Lambda}_q)$ 
11:  $S_{\mathcal{I}} \leftarrow \text{SpectralAssociativeMemory}(\mathcal{A}(\mathcal{I}))$ 
12: // Phase 2 — Online inference
13: Initialise transformer FFN / cross-attention from  $S_{\mathcal{I}}$ 
14: for each inference query  $\mathbf{q}$  do
15:   Retrieve via eq. (21); optionally update via eq. (22)
16: end for
```

---

**No runtime eigensolves.** Because  $(\mathbf{U}_q, \boldsymbol{\Lambda}_q)$  are frozen constants in Phase 1, *no Laplacian is computed or inverted at each training step*. All  $O(N^3)$  eigendecomposition cost is paid once in offline preprocessing.

**ELBO Bayes factor for index selection.** Training VDT on competing indices  $\mathcal{I}_1, \mathcal{I}_2$  yields ELBO estimates  $\mathcal{L}(\mathcal{I}_1), \mathcal{L}(\mathcal{I}_2)$ . The Bayes factor

$$\text{BF}_{12} = \exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2)) \quad (23)$$

provides a principled comparison of index constructions: an index whose spectral geometry explains the data covariance better will achieve a higher ELBO [Lotfi et al., 2022].

## 8 Stability Analysis

VDT inherits the full stability hierarchy from `docs/v2/04-stability.md`. Two additional conditions apply.

**Sample-graph Laplacian feedback.** When the Laplacian-precision latent KL (design option) is enabled,  $L_s$  is built from the batch of latent means  $\mu_z$ , creating a feedback loop. Prevention: apply stop-gradient on  $L_s$  construction; or use a frozen sample Laplacian refreshed every  $T_{\text{refresh}}$  epochs.

**Spectral-basis KL conditioning.** If  $\lambda_1 \approx 0$  (Fiedler vector near zero), the prior on the first mode becomes nearly flat, increasing gradient variance. Apply a floor  $\lambda_k \leftarrow \max(\lambda_k, \epsilon)$  with  $\epsilon = 10^{-3}$  in all KL computations (implemented in `vdt/spectral.py` as `eigvals.clamp(min=1e-6)`).

**Mode weight collapse.** High-frequency modes may collapse to  $\omega_k \rightarrow 0$  if  $\tau\lambda_k$  is large. Monitor active mode count  $|\{k : \mathbb{E}[\omega_k] > \delta\}|$ ; enforce a soft floor on the Gamma shape  $a_k \geq a_{\min}$ .

## 9 Benchmark Metrics

`evaluate()` in `vdt/metrics.py` computes the eight metrics in Table 3. The ELBO Bayes factor enables ArrowSpace index selection from data alone.

Table 3: VDT benchmark metrics.

| Metric                        | Computation                                                                                 | Measures                              |
|-------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------|
| Reconstruction MSE            | $\mathbb{E}_{\text{test}}[\ \mathbf{x} - \hat{\mathbf{x}}\ ^2]$                             | Quality of wiring-path reconstruction |
| KL spectral basis             | $\text{KL}(q(\mathbf{S})\ p(\mathbf{S} \mid \mathcal{I}))$                                  | Spectral alignment with index         |
| $\tau$ -mode KL               | $\text{KL}(q(\boldsymbol{\omega})\ p(\boldsymbol{\omega} \mid \tau, \boldsymbol{\Lambda}))$ | Frequency band selection              |
| Active mode count             | $\sum_k \mathbf{1}[\mathbb{E}[\omega_k] > 0.01]$                                            | Effective spectral bandwidth          |
| Memory retrieval SNR          | $d_k/N_{\text{stored}}$ (key orthogonality)                                                 | Associative memory quality            |
| ELBO Bayes factor             | $\exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2))$                             | ArrowSpace index comparison           |
| Linear probe accuracy         | Logistic regression on frozen $\mu$                                                         | Discriminative latent quality         |
| Spectral entropy $H(\Lambda)$ | $H(\text{normalised eigenvalues of } L(\mathbf{z}))$                                        | Wiring diversity                      |

## 10 Algorithm Tracks

The VDT framework supports six implementation tracks. The recommended sequence is  $6 \rightarrow 1 \rightarrow 4 \rightarrow 3$ .

- **Option 1 — Deterministic AE.** Replace  $J_{\text{freq}}$  with the soft  $\tau$ -mode KL; add `SpectralLoadingDecoder` as ablation. Backward-compatible:  $\beta \rightarrow 0$  recovers the deterministic limit.
- **Option 3 — Latent diffusion.** VDT provides the Stage 1 encoder; the spectral noise schedule  $a_\tau^{(k)} = \exp(-\tau\lambda_k t)$  matches the  $\Lambda_q^{-1}$  latent prior.

- **Option 4 — Variational Laplace AE.** VDT is the full MC-ELBO version; Laplace covariance  $(H_{\text{recon}} + \Lambda_m)^{-1}$  is the deterministic approximation to the spectral-precision posterior.
- **Option 6 — Classifier/Reasoner.** The signed density matrix  $\varrho_t = \varrho_t^+ - \varrho_t^-$  from the `VibrationalStateBlock` backbone can be used to initialise  $S_{\mathcal{I}}$ , grounding associative memory in reasoning-time spectral structure.

## 11 Relationship to Prior Work

**Graph VAEs.** Kipf and Welling [2016] encode graph structure into a VAE but do not impose Laplacian spectral priors on latent codes or loading matrices. VDT makes the spectral structure explicit and variational.

**Spectral graph convolution.** Graph spectral methods [Defferrard et al., 2016] use Laplacian eigenvectors as filter bases; VDT places this basis *in the generative model* as the loading matrix parametrisation.

**Laplacian priors.** Rao et al. [2008] and Slawski and Hein [2015] use Laplacian quadratic-form priors for Bayesian variable selection; VDT applies this to both the loading matrix  $\mathbf{S}$  and (optionally) the latent code  $\mathbf{z}$ .

**Hopfield / associative memory.** Ramsauer et al. [2021] extended Hopfield networks to continuous attractors; Geva et al. [2021] showed that transformer FFN layers are key-value memories. VDT unifies these by seeding the memory with spectral artefacts derived from vibrational graph dynamics.

**Vibrational mechanics.** The `VibrationalStateBlock` recurrence is a direct discretisation of the damped wave equation, connecting the architecture to Rayleigh’s Theory of Sound [?]. The positive/negative amplitude decomposition  $(\varrho_t^+, \varrho_t^-)$  is a real-valued analogue of the complex amplitude representation in standard quantum mechanics, as discussed in ? Chapter 9.

## 12 Discussion and Open Problems

**Posterior variance head for  $\mathbf{S}$ .** The current implementation approximates  $\log \text{var}_{S,kj} \approx \log(S_{kj}^2 + \epsilon)$ . Adding a dedicated variance head in `SpectralLoadingDecoder` would give a proper amortised Gaussian posterior for  $\mathbf{S}$  and is a prioritised open task.

**Laplacian-precision latent KL.** PR #35 removed this term from the default ELBO after finding that the feedback loop between  $\mu_z$  and the dynamically-constructed  $L_s$  required careful stop-gradient discipline and added training instability. A frozen sample Laplacian refreshed every  $T$  epochs may resolve this; experiments are ongoing.

**Nonlinear value function.** The current spectral artefact uses a linear decoder approximation  $d_\theta(\hat{\mathbf{w}}_k) \approx \hat{\mathbf{w}}_k$  for constructing  $S_{\mathcal{I}}$ . Using the full `DiffusionDecoder` response at each spectral direction would produce a proper key-value Hopfield structure and is left as future work.

**Iterative eigensolver.** For large graphs ( $N \gg 10^3$ ), truncated LOBPCG or ARPACK can replace full `eigh` in the offline phase, reducing the preprocessing cost from  $O(N^3)$  to  $O(N \cdot q)$ .

**Complex-number-free quantum analogy.** A core motivation for the signed density matrix  $\varrho_t = \varrho_t^+ - \varrho_t^-$  is to represent quantum-like superposition without complex numbers, following the observation in ? that negative probability amplitudes already capture the essential interference phenomena of quantum mechanics. The graph Laplacian in feature space serves as the minimal structure that admits this representation: its eigenmodes carry the role of quantum basis states, and the signed weights  $\omega_k^+ - \omega_k^-$  play the role of real-valued amplitudes. Formalising this correspondence is an open theoretical question.

## 13 Conclusion

The (VDT) reframes the Wiring Autoencoder as a principled Bayesian VAE in which the latent space and loading matrix are explicitly shaped by Laplacian spectral geometry provided by an ArrowSpace index. The three-term ELBO (eq. (11)) replaces the hard  $J_{\text{freq}}$  penalty with interpretable variational KL terms grounded in vibrational physics; the two-phase offline/online architecture eliminates runtime eigensolves; and the spectral artefact  $\mathcal{A}(\mathcal{I})$  provides a principled initialiser for transformer associative memory whose keys are the natural resonances of the data graph. ELBO Bayes factors make ArrowSpace index selection a first-class modelling decision. The signed density matrix  $\varrho_t = \varrho_t^+ - \varrho_t^-$  maintained throughout the `VibrationalStateBlock` recurrence establishes a real-valued analogue of quantum amplitude representation, connecting the architecture to both Rayleigh’s theory of vibrating systems and the foundational question of whether negative probability amplitudes suffice to recover quantum-like behaviour without complex numbers.

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